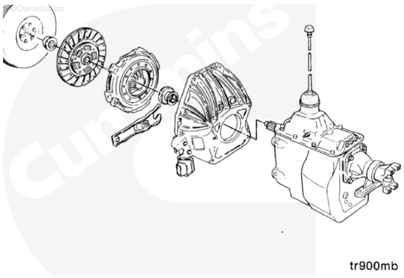
Trainings

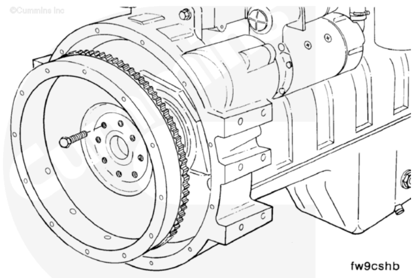
Remove

Remove the transmission, clutch, and all related components (if equipped). Refer to the manufacturer's instructions.



⚠ WARNING ⚠

The component weighs 23 kg [51 lb] or more. To avoid personal injury, use a hoist or get assistance to lift the component.

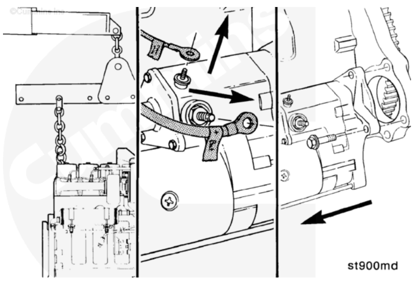


Remove the flywheel/ring gear assembly. Refer to Procedure 016-005 (/qs3/pubsys2/xml/en/procedures/100/100-016-005.html).

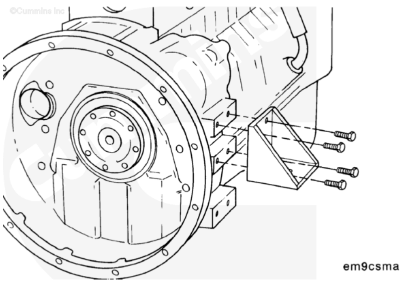
Adequately support the engine to prevent damage.

Disconnect the battery cables.

Remove the starter motor.

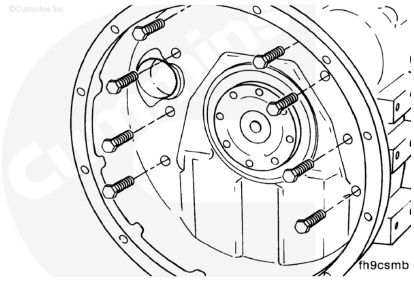


Remove the capscrews and both rear engine mounts.



⚠ WARNING ⚠

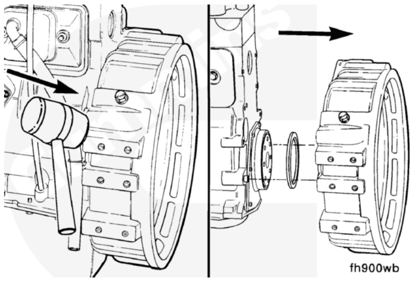
The component weighs 23 kg [51 lb] or more. To avoid personal injury, use a hoist or get assistance to lift the component.



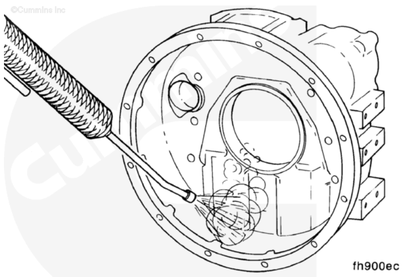
While supporting the flywheel housing, remove the mounting capscrews.

Use a rubber hammer to loosen the flywheel housing.

Remove the flywheel housing and rectangular seal.



Clean



⚠ WARNING ⚠

When using a steam cleaner, wear protective clothing, as well as safety glasses or a face shield. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles, as well as protective clothing, to avoid personal injury.

⚠ WARNING ⚠

Compressed air used for cleaning should not exceed 207 kPa [30 psi]. Use only with protective clothing, as well as goggles/shield, and gloves to avoid personal injury.

Use steam or solvent to clean the flywheel housing.

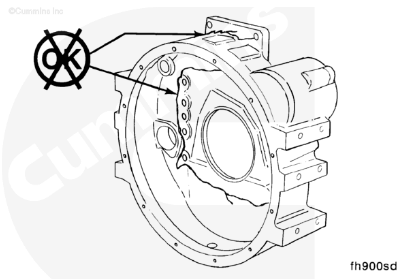
Dry with compressed air.

Inspect for Reuse

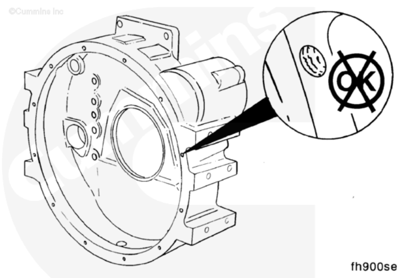
Inspect the flywheel housing for cracks, especially in the bolt pattern area.

Inspect all surfaces for nicks, burrs, or cracks.

Use fine crocus cloth to remove small nicks and burrs.



Inspect for damaged threads commonly caused by cross-threaded capscrews or installing an incorrect capscrew. Helicoils are available to repair damaged threads.

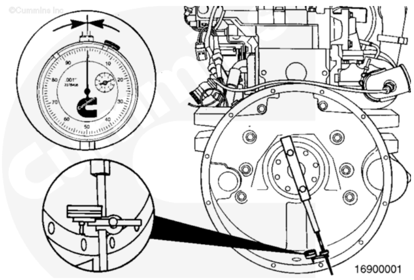


Measure

Bore Alignment

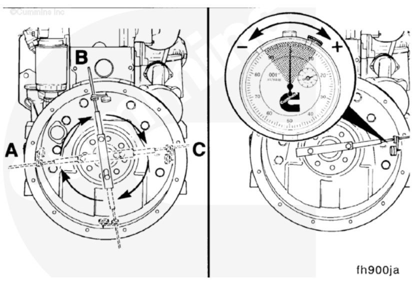
Attach the dial indicator gauge, Part No. 3376050, to the crankshaft. The dial indicator can be mounted by any method that holds the extension bar of the indicator rigid, so it does **not** sag. If the bar sags or the indicator slips, the readings obtained will **not** be accurate.

Position the indicator in the six-o'clock position, and zero the gauge.



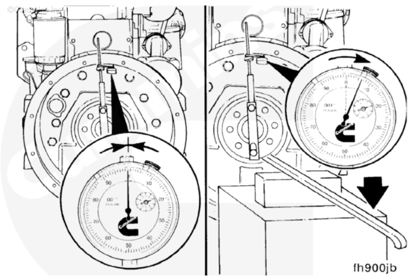
Slowly rotate the crankshaft. Record the readings obtained at the nine-o'clock, twelve-o'clock, and three-o'clock positions as [a], [b], and [c] in the concentricity work sheet. Recheck zero at the six-o'clock position.

The values for (a), (b), and (c) could be positive or negative. Refer to the accompanying figure to determine the correct sign when recording these values.



⚠ CAUTION ⚠

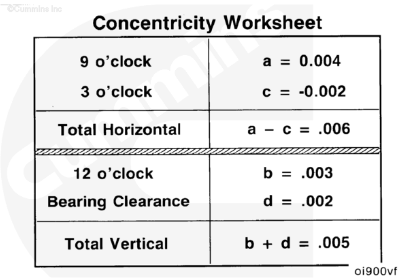
Do not force the crankshaft beyond the point where the bearing clearance has been removed. Do not pry against the flywheel housing. These actions could cause false bearing clearance readings.



Rotate the crankshaft until the dial indicator is at the twelve o'clock position and zero the gauge.

Using a pry bar, raise the rear of the crankshaft to its upper limit. Record the value as (d) in the concentricity work sheet. This is the vertical bearing clearance adjustment and will **always** be positive.

Using the concentricity work sheet, determine the values for the “total vertical” and “total horizontal” values.



Concentricity Worksheet

9 o'clock	a = 0.004
3 o'clock	c = -0.002
Total Horizontal	a − c = .006
12 o'clock	b = .003
Bearing Clearance	d = .002
Total Vertical	b + d = .005

oi900vf

The total horizontal is equal to the nine-o'clock reading (a), minus the three-o'clock reading (c).

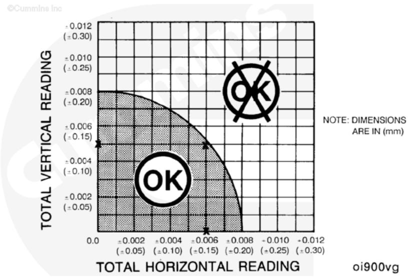
The total vertical is equal to the twelve-o'clock reading (b), plus the bearing clearance (d).

Example:

Six o'clock = ref = 0, Nine o'clock = (a) = 0.004, Twelve o'clock = (b) = 0.003, Three o'clock = (c) = -0.002

Using the work sheet and the numbers from the example, the total horizontal value equals 0.006 and the total vertical value equals 0.005.

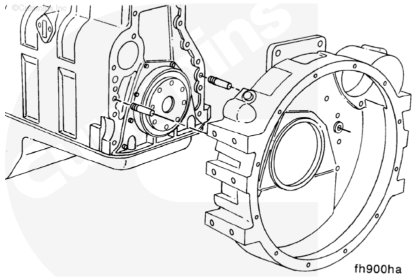
Mark the total horizontal value on the horizontal side of the chart and the total vertical on the vertical side of the chart.



Using a straightedge, find the intersection point of the total horizontal and total vertical values. The intersection point **must** fall within the shaded area for the flywheel housing concentricity to be within specification.

Using the total horizontal and total vertical values from the previous example, the intersection point falls within the shaded area. Therefore, the flywheel housing concentricity is within specification.

If the intersection point falls outside the shaded area, the ring dowels **must** be removed and the housing repositioned.



Note : The ring dowels are not required to maintain concentricity of the housing; the clamping force of the capscrews holds the housing in place.

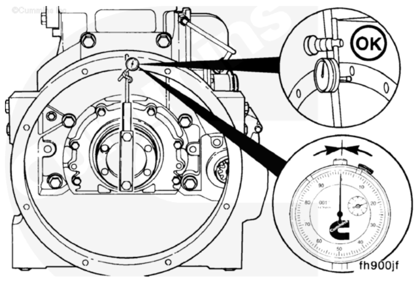
After the ring dowels are discarded, install the flywheel housing on the engine.

To position the housing, tighten the capscrews enough to hold the flywheel housing in place, but loose enough to allow small movement when struck lightly with a mallet.

Recheck the concentricity. When concentricity is within specification, tighten the capscrews to the specified torque value.

⚠ CAUTION ⚠

The dial indicator tip must not enter the capscrew holes, or the gauge will be damaged.

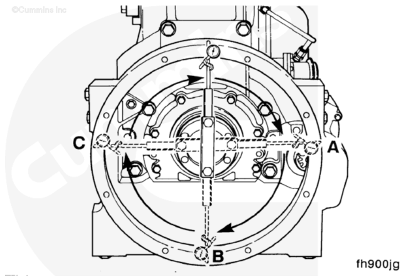


Face Alignment

Install the dial indicator gauge, Part No. 3376050, as illustrated.

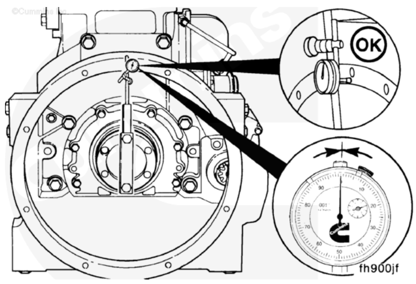
Note : The extension bar for the indicator must be rigid for an accurate reading. It must not sag. Position the indicator at the twelve-o'clock position. Adjust the dial until the needle points to zero.

Slowly rotate the crankshaft. Record the readings at the three-o'clock, six-o'clock, and nine-o'clock positions.



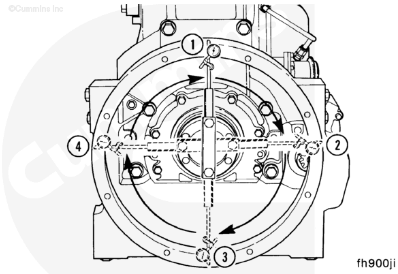
Note : The crankshaft must be pushed toward the front of the engine to remove the crankshaft end clearance each time a position is measured.

Continue to rotate the crankshaft until the indicator is at the twelve-o'clock position. Check the indicator to make sure the needle points to zero. If it does **not**, the readings will be incorrect.



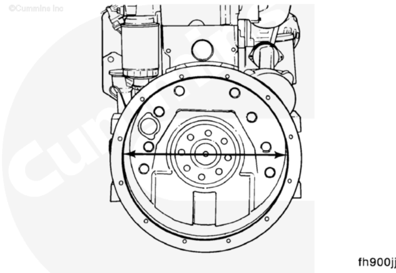
Determine the total indicator reading (TIR).

Example:	**mm**	**in**
12 o'clock	0.00	0.000
3 o'clock	+0.08	+0.003
6 o'clock	-0.05	-0.002
9 o'clock	+0.08	+0.003
Equals TIR	0.13	0.005



The maximum allowable total indicator reading (TIR) is determined by the diameter of the housing bore. If out of specifications, replace the housing.

SAE No.	**Bore Diameter**	**TIR**		
	mm	**in**	**mm**	**in**
2	447.68 to 447.80	17.625 to 17.30	0.20	0.008
3	409.58 to 409.70	16.125 to 16.130	0.20	0.008

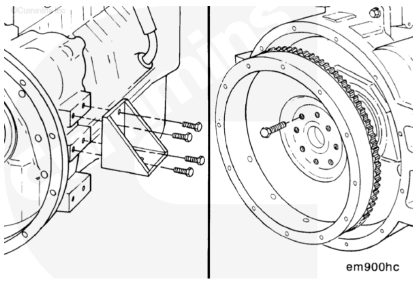


Install

Install both rear engine mounts.

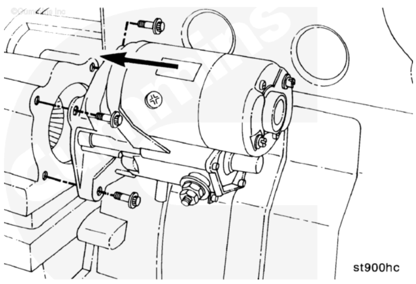
Install the flywheel and clutch (if equipped). Refer to the manufacturer's instructions.

Install the transmission and related components. Refer to the manufacturer's instructions.



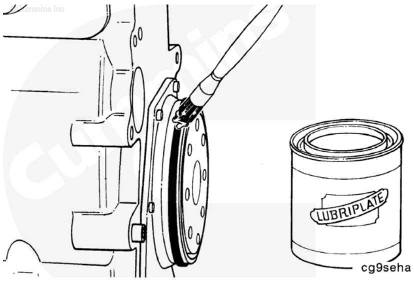
Install the starter motor. Refer to Procedure 013-020 (/qs3/pubsys2/xml/en/procedures/100/100-013-020.html).

Connect the battery cables. Refer to Procedure 013-009 (/qs3/pubsys2/xml/en/procedures/100/100-013-009.html).



Dry Clutch Application

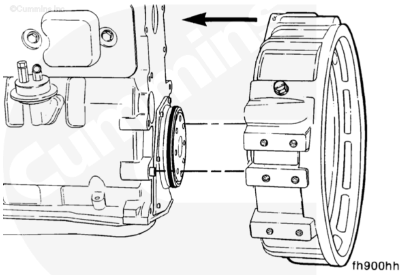
Install rectangular seal, and lubricate with Lubriplate™ 105, or equivalent.



Inspect the rear face of the cylinder block and flywheel housing mounting surface for cleanliness and raised nicks or burrs.

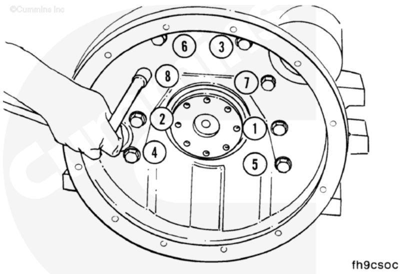
Install the flywheel housing over the two ring dowels.

Note : Be sure the sealing ring is not damaged during installation.



Tighten the flywheel housing capscrews in the sequence shown.

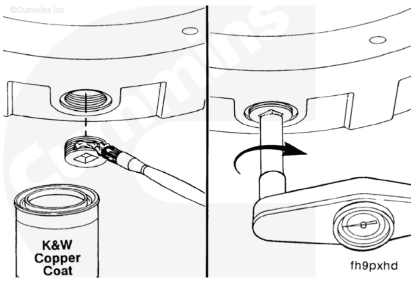
Torque Value: 77 n•m [57 ft-lb]



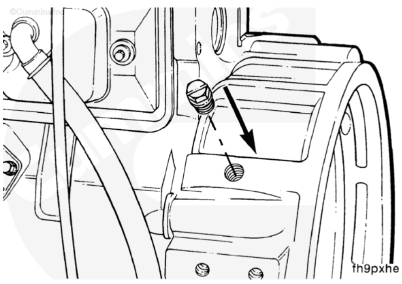
Wet Clutch Application

Perform all the steps in the procedure for dry clutch installation, in addition to the following:

- Coat the flywheel housing drain plug with pipe sealant, and install in the hole in the bottom of the flywheel housing.
- Tighten the plug.
- Refer to the pipe plug torque values in Section 17 for different plug sizes.



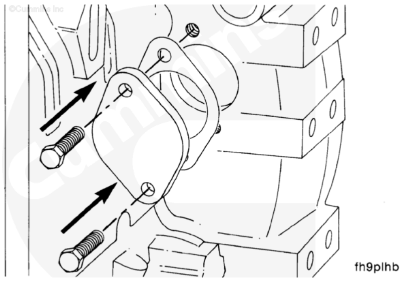
Install the plastic plug in the tachometer drive access hole.

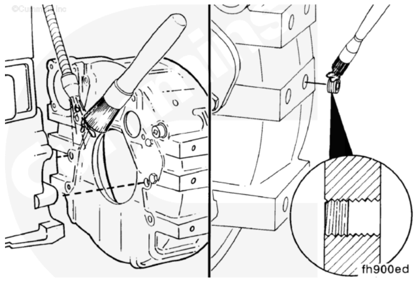


Install the access plate and new gasket.

Install the capscrews and tighten.

Torque Value: 24 n•m [18 ft-lb]





⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

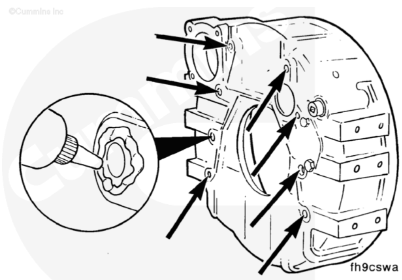
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.

Thoroughly clean the flywheel housing and cylinder block mating surfaces. These surfaces **must** be clean of oil and debris.

Note : The capscrew holes on the mounting pads are drilled through. Coat set screws with Loctite 277 and install into holes.

mm		in
3.00	MAX	0.118

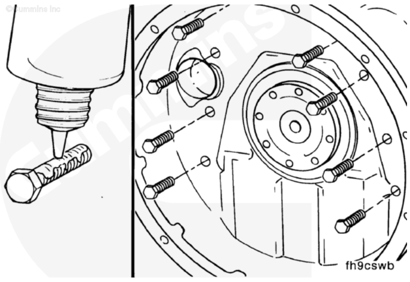
Apply a continuous bead of Three-Bond™ around all capscrew holes on the mounting surface of the flywheel housing.



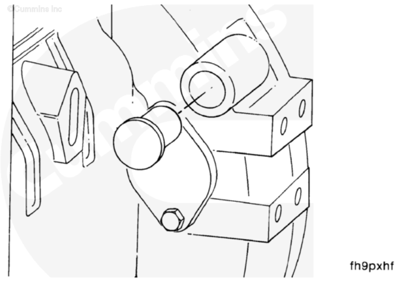
Coat the threads of the mounting capscrews with Loctite 277.

Install and tighten the capscrews.

Torque Value: 77 n•m [57 ft-lb]



Install the plug into the barring gear hole.



Last Modified: 21-Jul-2003